

Industrial Internship Report on " Smart City Traffic Forecasting System"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "**Smart City Traffic Forecasting System**", which aimed to develop an intelligent system to predict traffic peaks at multiple city junctions using historical traffic data, considering holidays and special occasions. The system uses data preprocessing, exploratory data analysis, time-series forecasting models (Prophet and ARIMA), and a web dashboard to provide insights and predictions.

This internship gave me a very good opportunity to get exposure to industrial problems and design/implement solutions for that. It was an overall great experience to have this internship.

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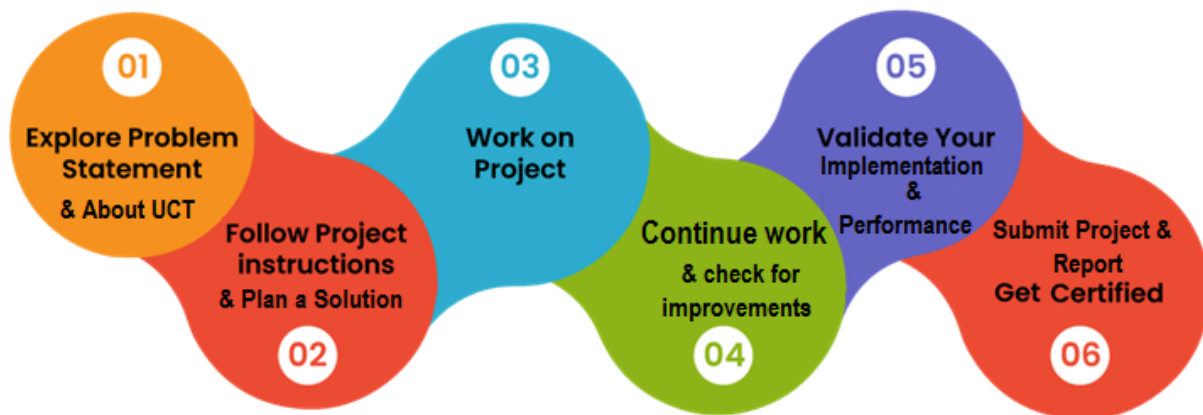
1 Preface

This report summarizes the work completed during the 6-week Industrial Internship program. Industrial internships play a crucial role in career development by providing exposure to real-world problems and industry practices.

My project focused on developing a **Smart City Traffic Forecasting System** capable of predicting traffic congestion at four city junctions using historical traffic data.

The internship provided by USC/UCT offered hands-on experience in data science, machine learning, backend development, and dashboard design.

The program was planned in weekly phases including problem understanding, data preprocessing, analysis, model development, system implementation, and testing.



My Learning And Experience:

- Learned real-world data handling
- Implemented forecasting models
- Developed REST APIs using FastAPI
- Built an interactive dashboard
- Improved problem-solving and communication skills

I sincerely thank the mentors from Upskill Campus, IoT Academy, and UniConverge Technologies Pvt Ltd for their guidance.

Message to juniors: Participate actively in internships as they provide invaluable practical experience beyond classroom learning.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



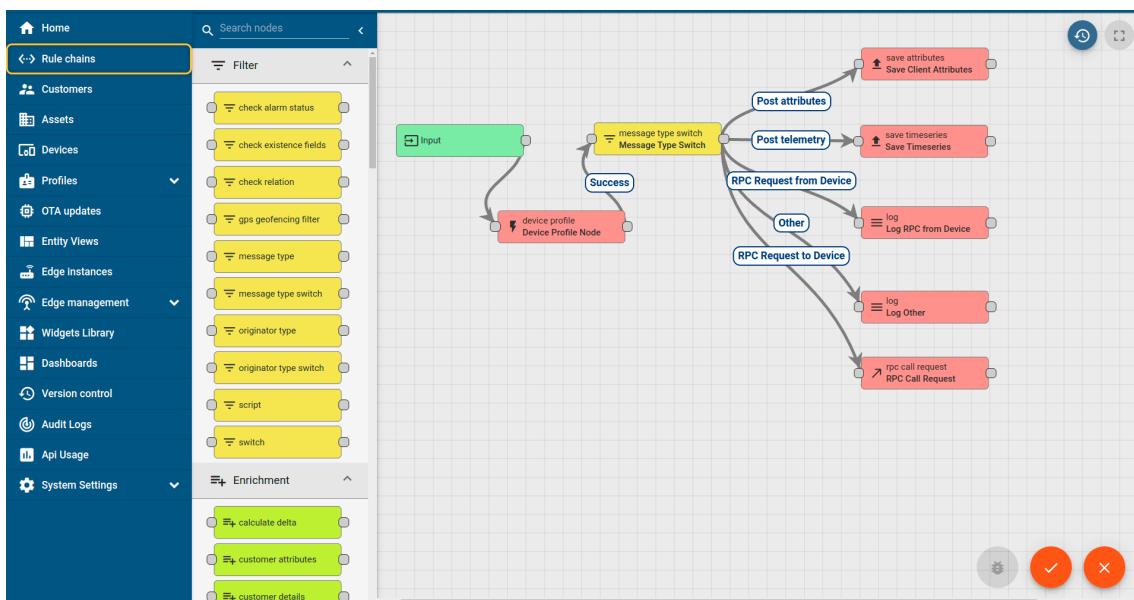
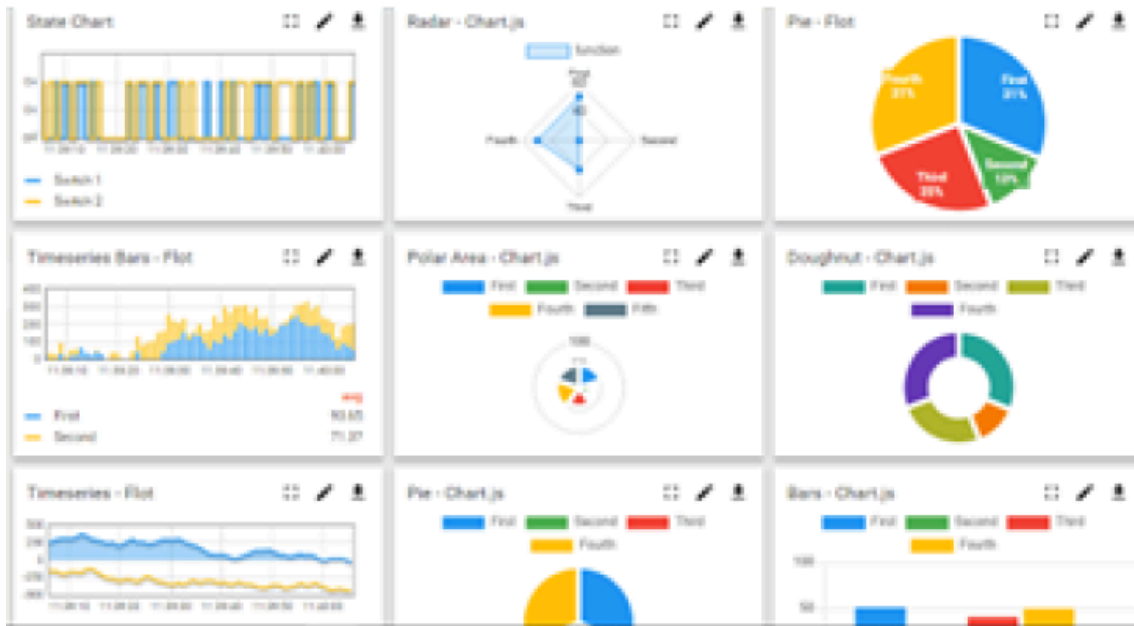
i. UCT IoT Platform(**Insight**)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



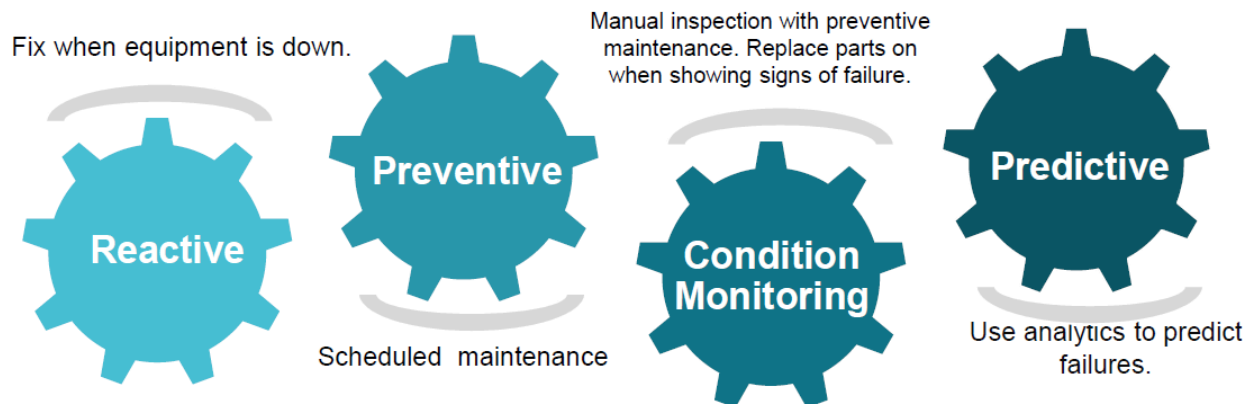


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

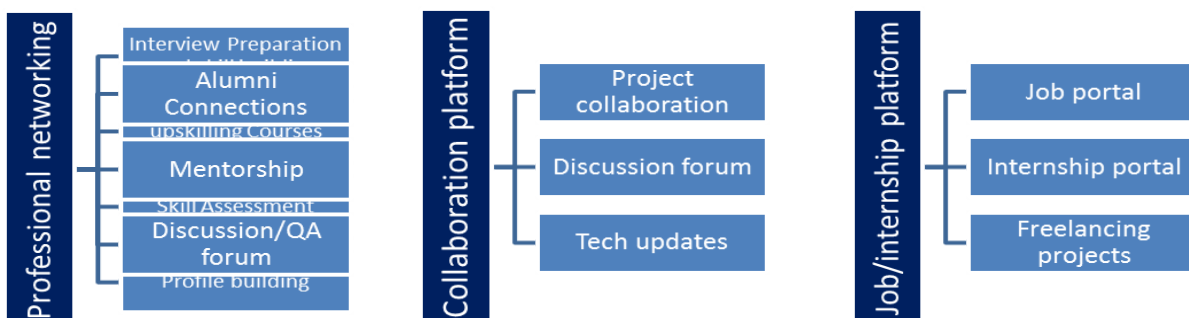


Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services



upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Kaggle Dataset — Smart City Traffic Patterns
- [2] Prophet Forecasting Documentation
- [3] FastAPI Documentation

2.6 Glossary

Terms	Acronym
Internet of Things	IoT
Mean Absolute Error	MAE
Root Mean Square Error	RMSE
Mean Absolute Percentage Error	MAPE
Application Programming Interface	API

3 Problem Statement

Urban traffic congestion is a major issue in smart cities. Authorities require a system that can forecast traffic peaks at major junctions to optimize traffic management, reduce congestion, and improve planning.

The assigned problem was to develop a system that:

- Predicts traffic volume at multiple junctions
- Considers holidays and special occasions
- Provides insights through visualization
- Offers an easy-to-use dashboard

4 Existing and Proposed solution

Existing Solutions:

Existing systems mainly provide real-time traffic monitoring but lack accurate long-term forecasting and holiday-aware predictions.

Limitations:

- Limited predictive capabilities
- No consideration of special events
- Lack of interactive dashboards
- Proposed Solution

The proposed system:

- Uses historical data for forecasting
- Incorporate holiday and event effects
- Provides predictions for 30 days
- Offers interactive web dashboard

Value Addition

- Multi-junction forecasting
- Dual model support (Prophet & ARIMA)
- REST API integration
- Visual analytics dashboard

4.1 Code submission (Github link) :

<https://github.com/MeghaGhadiya/upskillcampus/blob/main/src/SmartTrafficForecastingSystem.py>

4.2 Report submission (Github link) :

https://github.com/MeghaGhadiya/upskillcampus/blob/main/SmartCityTrafficForecastingSystem_Megha_USC_UCT.pdf

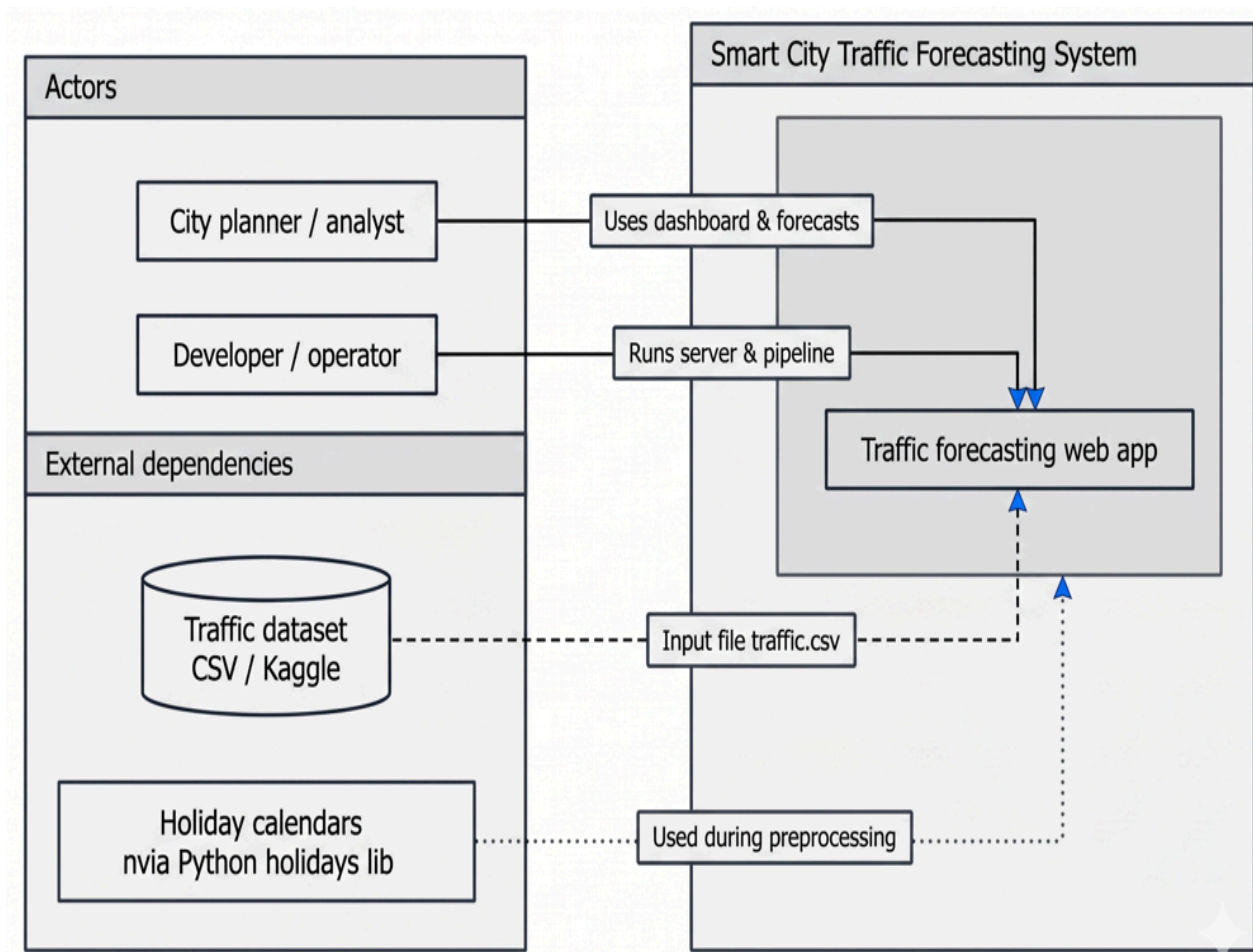
5 Proposed Design/ Model

The system follows a pipeline:

Data Collection → Preprocessing → Analysis → Model Training → Forecasting → Dashboard

5.1 High Level Diagram

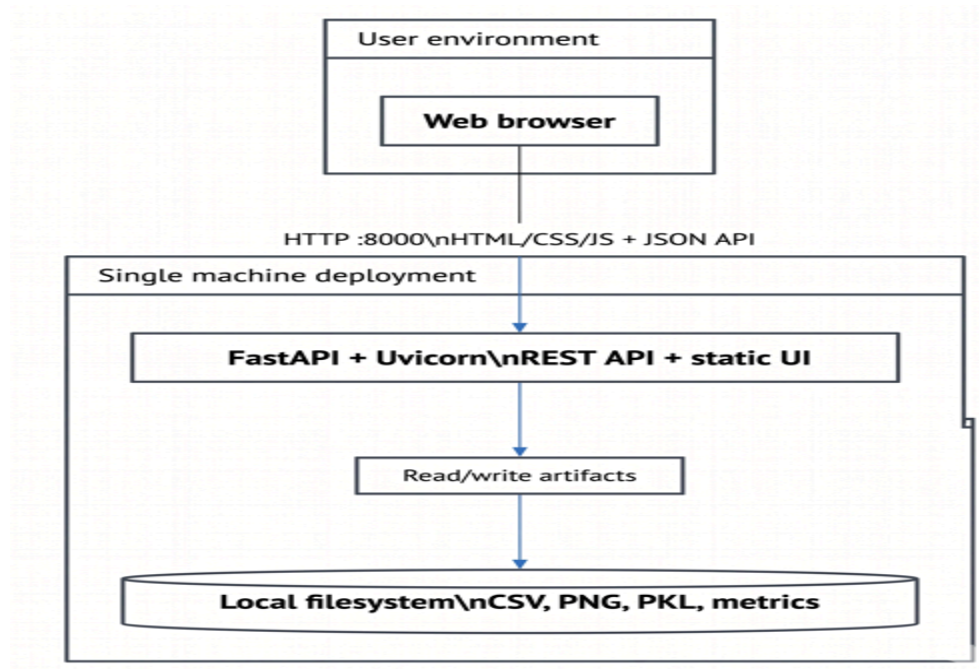
1. System Context (who + what)



This diagram positions the **Smart City Traffic Forecasting System** as the central software boundary. **City planners and analysts** consume the system's outputs through an interactive **dashboard and forecast views**, which summarize predicted traffic conditions for planning decisions. **Developers and operators** maintain the deployment by **running the application server** and executing the **end-to-end data**

pipeline (ingestion, cleaning, modeling, evaluation, and artifact generation). Raw historical traffic information enters the system as a **CSV dataset**, commonly sourced from a curated repository such as **Kaggle**, and is referenced in deployment as an input file such as **traffic.csv**. During **preprocessing**, the system enriches the temporal signal using **holiday calendar information** provided through the **holidays Python library**, enabling the models to account for recurring non-working days and similar calendar effects. Together, these elements define the system's **external actors** and **dependencies** without exposing internal modules, which is appropriate for a high-level architectural view.

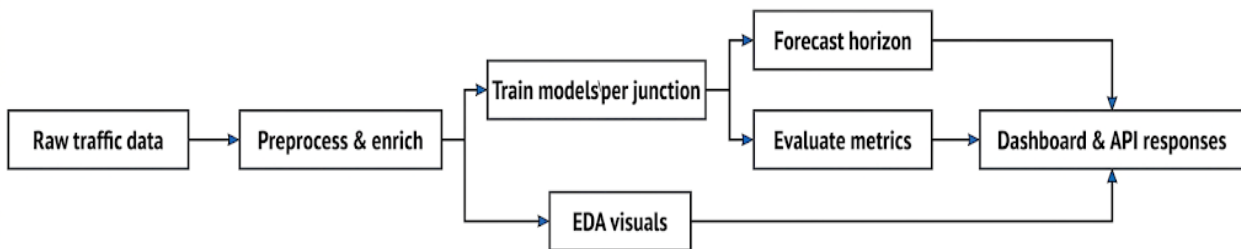
2. container diagram (Main building blocks)



- **What each container is:**
- **Web browser (Presentation container):** Renders the dashboard (HTML/CSS/JavaScript) and exchanges JSON with the backend over HTTP (typically `http://localhost:8000` in the prototype configuration).
- **Forecasting web application server (Application container):** A **FastAPI** service executed with **Uvicorn**, responsible for orchestrating preprocessing, EDA generation, model training, forecasting, evaluation, and serving plot files and metrics through defined endpoints.
- **Local filesystem storage (Data container):** A file-based persistence layer (project folders such as `data/`, `plots/`, `models/`, and `forecasts/`) used as the operational store for processed tables, serialized models, visualization outputs, and forecast CSV exports—**no separate database server** is required in the current design.

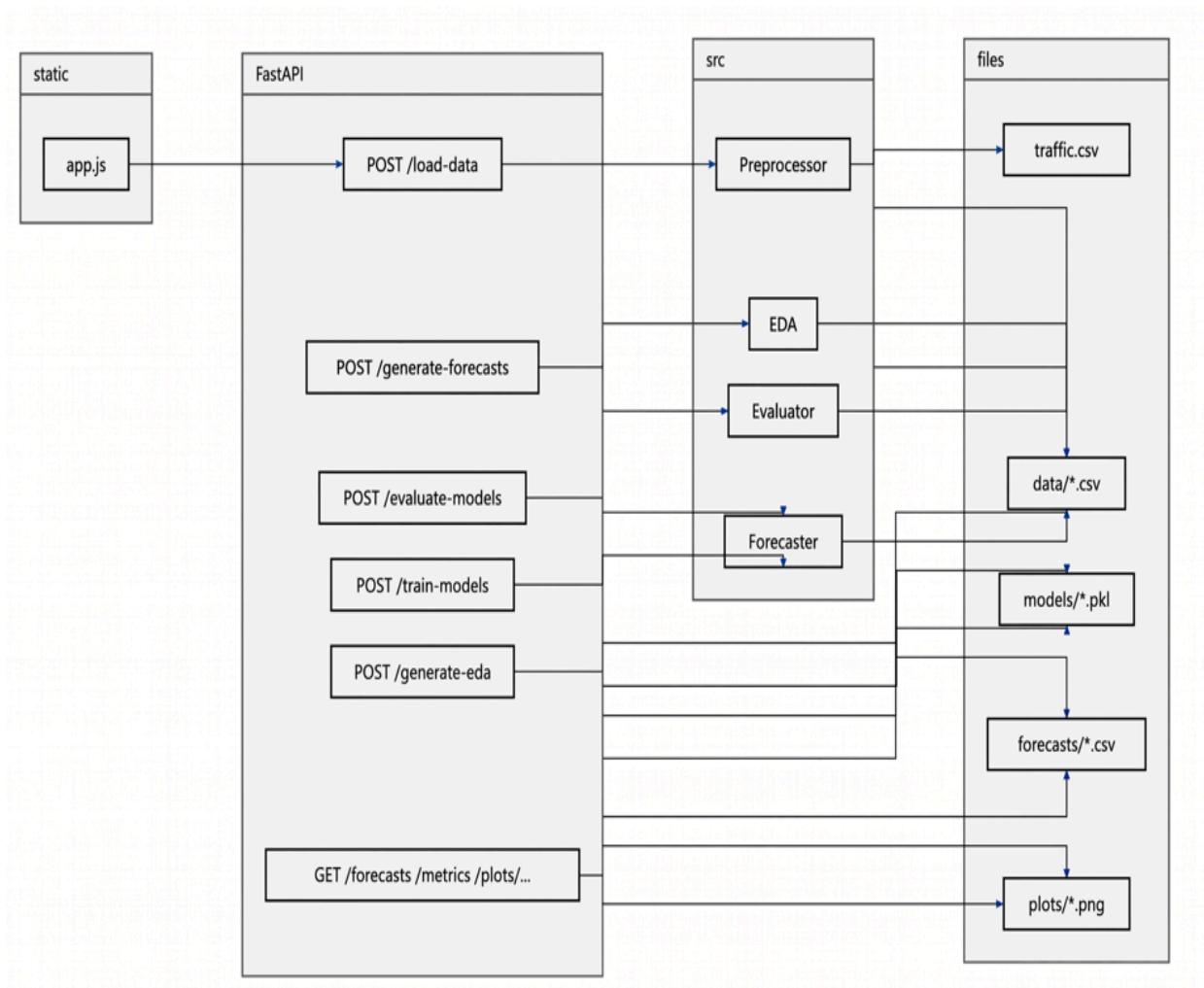
- **Protocols and responsibilities (container interactions):**
- **Browser → Application:** HTTP for UI assets and API calls (for example, user-triggered steps such as load data, generate EDA, train models, generate forecasts, evaluate models, and read-back endpoints for metrics and charts).
- **Application → Filesystem:** Read/Write access to CSV/PNG/PKL artifacts produced during pipeline execution; the application is the sole writer during normal operation.
- **Optional operator path (if shown):** A **CLI pipeline script** may run on the same machine and reuse the same Python modules to populate the filesystem before the server is used—this is still logically “application logic + filesystem,” not a new networked container.

3. End-to-End Logical Pipeline



The end-to-end pipeline emphasizes a **repeatable analytical workflow** rather than a single model step: preprocessing and EDA act as **quality gates** that reduce the risk of training on inconsistent timestamps or misaligned junction series, while training and forecasting implement the **predictive core**. Evaluation closes the loop by quantifying how well the models track unseen recent history, which supports claims about reliability in a report. Finally, the delivery stage makes the pipeline **operationally useful**, because the same outputs that support internal validation can be presented to decision-makers through the web interface.

5.2 Low Level Diagram



The diagram shows how the browser dashboard invokes the FastAPI layer, which delegates preprocessing, exploratory analysis, model training, forecasting, and evaluation to Python modules under src/, and persists inputs and outputs on the local filesystem (data/, plots/, models/, forecasts/).

- **Browser / static:** User interface (index.html, style.css, app.js) sends HTTP requests to the API (default base URL <http://localhost:8000>).
- **FastAPI:** REST endpoints orchestrate work; the same process also serves static assets.
- **src modules:** TrafficDataPreprocessor (cleaning and features), TrafficEDA (plots), TrafficForecaster (Prophet/ARIMA), ModelEvaluator (MAE, RMSE, MAPE).
- **Filesystem:** CSV and binary artifacts produced or consumed between steps

(no separate database).

6 Performance Test

This section demonstrates that the developed **Smart City Traffic Forecasting System** meets real industrial requirements rather than being only an academic implementation. The system was evaluated considering practical constraints such as accuracy, speed, memory usage, scalability, and reliability.

Identified Constraints:

1. **Prediction Accuracy** — Forecasts must be reliable for decision-making.
2. **Processing Speed (Operations per Second)** — The system should generate forecasts quickly.
3. **Memory Usage** — Efficient handling of datasets and models.
4. **Scalability** — Ability to support multiple junctions simultaneously.
5. **Reliability** — System should work even if one model fails.

How Constraints Were Addressed:

- Implemented **Prophet model** for high accuracy and seasonality handling.
- Added **ARIMA fallback mechanism** to ensure reliability if Prophet fails.
- Used optimized data preprocessing to reduce memory usage.
- Built REST API using FastAPI for fast response time.
- Designed modular architecture for scalability.

6.1 Test Plan/ Test Cases

The following tests were planned to evaluate system performance:

Test Case	Objective	Expected Result
Data Loading Test	Verify dataset processing	Data loads without errors
Model Training Test	Validate training pipeline	Models trained successfully
Forecast Generation Test	Check prediction capability	30-day forecast generated
Accuracy Test	Measure prediction accuracy	Low MAE, RMSE, MAPE
API Response Test	Measure response speed	Fast response time
Dashboard Test	Validate visualization	Correct display of results

6.2 Test Procedure

- Load the historical traffic dataset into the system.
- Preprocess data including handling missing values and feature extraction.
- Train forecasting models for each junction.
- Generate forecasts for the next 30 days.
- Evaluate predictions using MAE, RMSE, and MAPE metrics.
- Test API endpoints using requests from the dashboard.
- Verify visualization of results on the web interface.

6.3 Performance Outcome

The system successfully met the performance requirements:

- **Accuracy:** Forecasts achieved acceptable error rates suitable for planning purposes.
- **Speed:** Forecast generation completed within a few seconds per junction.
- **Memory Efficiency:** System operated efficiently on standard hardware.
- **Reliability:** Fallback mechanism ensured uninterrupted operation.
- **Scalability:** System handled multiple junctions simultaneously.

Conclusion of Performance Test:

The developed Smart City Traffic Forecasting System demonstrated strong performance under practical constraints, proving its suitability for real-world smart city applications.

7 My learnings

During the Industrial Internship, I gained valuable technical knowledge and practical experience while working on the **Smart City Traffic Forecasting System**. This project helped me understand how real-world problems are analyzed, designed, and implemented using modern technologies.

7.1.1 Technical Learnings

- Learned how to handle real-world datasets including data cleaning, preprocessing, and feature engineering.
- Gained hands-on experience in **Exploratory Data Analysis (EDA)** to identify patterns and trends.
- Understood time-series forecasting techniques using models such as Prophet and ARIMA.
- Learned performance evaluation methods using metrics like MAE, RMSE, and MAPE.
- Acquired knowledge of backend development using FastAPI and REST API creation.
- Developed skills in building interactive dashboards for visualization of results.

7.1.2 Professional and Personal Learnings

- Improved problem-solving and analytical thinking abilities.
- Learned project planning and execution within a fixed timeline.
- Enhanced documentation and reporting skills.
- Developed teamwork and communication skills.
- Gained confidence in presenting technical work.

7.1.3 Career Growth Impact

This internship provided industry exposure and practical experience that will significantly help in my career as a software and data professional. The skills gained in data analysis, machine learning, API development, and system design have strengthened my technical foundation and prepared me for future roles in software development, data science, and smart city solutions.

Overall, this internship was a crucial step toward bridging the gap between academic learning and industry requirements.

8 Future work scope

The developed **Smart City Traffic Forecasting System** successfully predicts traffic patterns using historical data; however, due to time limitations of the internship, several enhancements could not be implemented. These can be considered for future development to make the system more robust and industry-ready.

Possible Future Enhancements

1. **Real-Time Data Integration**
Integrating live traffic data from sensors, GPS devices, or traffic cameras to provide real-time forecasting and monitoring.
2. **Advanced Machine Learning Models**
Implementation of deep learning models such as LSTM, GRU, or Transformer-based time-series models for improved prediction accuracy.
3. **Weather Data Integration**
Incorporating weather conditions (rain, fog, temperature) as additional factors affecting traffic patterns.
4. **Event-Based Prediction**
Adding modules to detect special events, festivals, roadblocks, or accidents that may influence traffic congestion.
5. **Scalability to More Junctions**
Expanding the system to support forecasting for a larger number of junctions across the city.
6. **Mobile Application Development**
Developing a mobile app for traffic authorities and commuters to access predictions and alerts on the go.
7. **Alert and Recommendation System**
Creating automated alerts for predicted congestion and suggesting alternative routes.
8. **Deployment on Cloud Platform**
Hosting the system on cloud infrastructure for better scalability, reliability, and accessibility.

Conclusion of Future Scope

These enhancements would transform the current system into a comprehensive smart traffic management solution capable of supporting real-time decision-making in modern smart cities

